

Cloud Computing

Mock Exam 1 | Course ID: noc26_cs55

Q1. What does XaaS stand for in cloud computing?

- A) eXtra as a Service
- B) Anything as a Service
- C) XML as a Service
- D) eXecution as a Service

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Easy

Q2. What is the primary purpose of a Service Level Agreement (SLA) in cloud computing?

- A) To define the programming language for cloud applications.
- B) To formally contract the level of service between provider and consumer.
- C) To specify the hardware used in data centers.
- D) To manage internal network traffic.

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Easy

Q3. Which concept is referred to as the 'mother of the things' for cloud computing, providing fundamental building blocks like fault tolerance and scalability?

- A) Utility Computing
- B) Grid Computing
- C) Distributed Computing
- D) Cluster Computing

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Medium

Q4. How does Clouddonomics influence strategic decision-making for organizations considering cloud adoption?

- A) It primarily focuses on reducing the number of physical servers owned by an organization.
- B) It guides the 'rent vs. buy' decision for computational resources, determining if hiring resources is more economical than purchasing and maintaining infrastructure.
- C) It mandates the use of open-source software to minimize licensing costs.
- D) It dictates the specific programming languages to be used for cloud-native applications.

Source: Lecture_W1L19_Lecture_1__Cloud_Computing__Overview | Difficulty: Hard

Q5. Which computing paradigm introduced the concept of treating computing as a metered service, where users are charged based on specific usage rather than a flat rate?

- A) Distributed Systems
- B) Grid Computing
- C) Cluster Computing
- D) Utility Computing

Source: Lecture_W1L20_Lecture_2__Cloud_Computing__Overview__contd__ | Difficulty: Easy

Q6. How does Cluster Computing primarily differ from Grid Computing in terms of coupling and primary goal?

- A) Cluster computing is loosely coupled and focuses on harnessing underutilized resources, while grid computing is tightly coupled and aims for high performance.
- B) Cluster computing is tightly coupled for high performance and availability, while grid computing is loosely coupled for virtualizing resources from multiple locations.
- C) Cluster computing focuses on scientific collaboration, while grid computing focuses on load balancing.
- D) Cluster computing uses proprietary hardware, while grid computing uses commodity hardware.

Q7. The NIST definition of cloud computing emphasizes resources that can be 'rapidly provisioned and released with minimal management effort.' Which earlier computing paradigm laid the conceptual groundwork for this on-demand, utility-like access by abstracting infrastructure complexity from the user?

- A) Distributed Systems
- B) Grid Computing
- C) Utility Computing
- D) Cluster Computing

Source: Lecture_W1L20_Lecture_2_Cloud_Computing____Overview____contd____ | Difficulty: Hard

Q8. Which NIST essential characteristic of cloud computing allows consumers to unilaterally provision computing capabilities as needed without requiring human interaction with each service provider?

- A) Broad Network Access
- B) Resource Pooling
- C) On-Demand Self-Service
- D) Measured Service

Source: Lecture_W1L21_Lecture_3_Cloud_Computing____Introduction | Difficulty: Easy

Q9. In the cloud service model hierarchy, which layer provides ready-to-use software applications delivered over the web, where the user simply consumes the service?

- A) Infrastructure as a Service (IaaS)
- B) Platform as a Service (PaaS)
- C) Software as a Service (SaaS)
- D) Backend as a Service (BaaS)

Source: Lecture_W1L21_Lecture_3_Cloud_Computing____Introduction | Difficulty: Easy

Q10. What is the primary role of a hypervisor (Virtual Machine Monitor) in cloud computing?

- A) To manage network routing between data centers.
- B) To create and run virtual machines (VMs) and manage the allocation of physical resources to each VM.
- C) To provide ready-to-use software applications over the internet.
- D) To define the legal structure and content of XML documents.

Source: Lecture_W1L21_Lecture_3_Cloud_Computing____Introduction | Difficulty: Medium

Q11. An organization needs to host a highly sensitive application with strict compliance requirements, but also wants to leverage a public cloud for non-sensitive, burstable workloads. Which cloud deployment model would be most suitable for this scenario?

- A) Public Cloud
- B) Private Cloud
- C) Community Cloud
- D) Hybrid Cloud

Source: Lecture_W1L21_Lecture_3_Cloud_Computing____Introduction | Difficulty: Medium

Q12. Explain how 'Resource Pooling' and 'Rapid Elasticity' are interconnected and enabled by virtualization to achieve cloud economics.

- A) Resource pooling ensures data is stored redundantly, while rapid elasticity allows for quick data recovery.
- B) Resource pooling aggregates diverse demands onto shared infrastructure, and rapid elasticity allows dynamic scaling of these pooled resources, both enabled by virtualization, to maximize utilization and align costs with needs.

- C) Resource pooling limits access to specific users, while rapid elasticity prevents unauthorized access to resources.
- D) Resource pooling dictates fixed resource allocation, while rapid elasticity allows for manual adjustments.

Source: Lecture_W1L21_Lecture_3__Cloud_Computing__Introduction | Difficulty: Hard

Q13. Which type of scaling involves adding more machines (nodes) to a resource pool to increase capacity, and is the primary method for achieving elasticity in the cloud?

- A) Vertical Scaling (Scale-Up)
- B) Horizontal Scaling (Scale-Out)
- C) Diagonal Scaling
- D) Elastic Scaling

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Easy

Q14. Why is 'loose coupling' considered essential for building scalable and resilient cloud systems?

- A) It ensures that all components are tightly integrated for maximum performance.
- B) It allows components to be updated, replaced, or scaled independently without causing system-wide failure.
- C) It reduces the need for network communication between services.
- D) It requires all services to be written in the same programming language.

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Medium

Q15. In the context of cloud architecture, explain how 'Vertical Stacks' like Management and Security differ from horizontal layers like IaaS or PaaS.

- A) Vertical stacks are confined to a single layer, while horizontal layers cut across the entire service stack.
- B) Vertical stacks represent cross-cutting concerns that permeate every layer of the architecture, from hardware to application, whereas horizontal layers are distinct, abstracted service models.
- C) Vertical stacks are only relevant in private clouds, while horizontal layers are for public clouds.
- D) Vertical stacks refer to adding more resources to a single machine, while horizontal layers refer to adding more machines.

Source: Lecture_W1L22_Lecture_4__Cloud_Computing_Architecture | Difficulty: Hard

Q16. What is the core technology that fundamentally differentiates cloud computing from the traditional client-server model, enabling multi-tenancy and dynamic resource allocation?

- A) Dedicated hardware
- B) Strong binding
- C) Virtualization
- D) Fixed IP addresses

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Easy

Q17. For which cloud service model is the provider responsible for the entire stack, from physical hardware to the application layer, with the user simply consuming the software?

- A) Infrastructure as a Service (IaaS)
- B) Platform as a Service (PaaS)
- C) Software as a Service (SaaS)
- D) Network as a Service (NaaS)

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Medium

Q18. A startup needs to quickly develop and deploy a new web application, minimize infrastructure management overhead, and reduce the risk of vendor lock-in as much as possible. Which cloud service model would be most suitable, and what is a key trade-off they should consider regarding vendor lock-in?

- A) IaaS; they gain full control but increase management overhead.
- B) PaaS; they abstract away OS and hardware, but might face vendor lock-in due to proprietary languages or tools.
- C) SaaS; they get ready-to-use software, but have no control over the application logic.
- D) Public Cloud; they get scalability, but have less security.

Source: Lecture_W1L23_Lecture_5__Cloud_Computing_Architecture__contd__ | Difficulty: Hard

Q19. Which cloud deployment model is provisioned for open use by the general public, owned and operated by a third-party provider, and offers maximum scalability?

- A) Private Cloud
- B) Community Cloud
- C) Public Cloud
- D) Hybrid Cloud

Source: Lecture_W2L28_Lecture_6__Cloud_Computing_Architecture__Deploymentm | Difficulty: Easy

Q20. What is a significant disadvantage of the Public Cloud model related to its multi-tenant nature?

- A) High upfront capital expenditure.
- B) Limited elasticity and scalability.
- C) Multi-tenancy risk, where workloads may share physical hardware with other tenants, introducing potential security and reliability risks.
- D) Complete control over the underlying infrastructure.

Source: Lecture_W2L28_Lecture_6__Cloud_Computing_Architecture__Deploymentm | Difficulty: Medium

Q21. An organization is considering a Hybrid Cloud deployment to leverage both on-premise control and public cloud scalability. What is a primary challenge they must address to ensure successful operation?

- A) Ensuring maximum physical security for the public cloud portion.
- B) Managing extreme complexity, interoperability, and data/application portability between different cloud infrastructures.
- C) Eliminating all capital expenditures for the private cloud component.
- D) Reducing network dependency for the public cloud.

Source: Lecture_W2L28_Lecture_6__Cloud_Computing_Architecture__Deploymentm | Difficulty: Medium

Q22. What is the software layer that creates and runs virtual machines (VMs) and manages the allocation of physical resources to each VM, often referred to as a Virtual Machine Monitor (VMM)?

- A) Operating System
- B) Application Software
- C) Hypervisor
- D) Middleware

Source: Lecture_W2L29_Lecture_7__Cloud_Computing_Virtualization | Difficulty: Easy

Q23. In an Infrastructure as a Service (IaaS) environment, who typically has control over the Guest OS, middleware, and application software running within a virtual machine?

- A) The Cloud Provider
- B) The Cloud Subscriber
- C) A third-party auditor
- D) The Hypervisor

Source: Lecture_W2L29_Lecture_7__Cloud_Computing_Virtualization | Difficulty: Easy

Q24. Compare and contrast Full Virtualization, Paravirtualization, and Hardware-Assisted Virtualization in terms of guest OS modification requirements and performance implications.

- A) Full virtualization requires a modified guest OS for high performance, while paravirtualization and hardware-assisted virtualization run unmodified guest OSs with lower performance.
- B) Full virtualization runs an unmodified guest OS with performance overhead, paravirtualization requires a modified guest OS for improved performance, and hardware-assisted virtualization runs an unmodified guest OS efficiently using processor extensions.
- C) Paravirtualization is the oldest method and the slowest, while full virtualization is the newest and fastest.
- D) Hardware-assisted virtualization is only used for Linux guest OSs, while full virtualization is for Windows.

Source: Lecture_W2L29_Lecture_7__Cloud_Computing__Virtualization | Difficulty: Hard

Q25. What is the primary purpose of XML (Extensible Markup Language) in the context of cloud computing and web services?

- A) To display data visually in web browsers.
- B) To structure and transport data for machine-to-machine communication.
- C) To define network protocols for data transfer.
- D) To manage hardware resources in a data center.

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Easy

Q26. What is the key difference between a 'well-formed' XML document and a 'valid' XML document?

- A) A well-formed document has correct syntax, while a valid document also conforms to a schema (like DTD or XSD).
- B) A well-formed document conforms to a schema, while a valid document only needs correct syntax.
- C) A well-formed document can be processed by any parser, while a valid document can only be processed by a validating parser.
- D) There is no difference; the terms are interchangeable.

Source: Lecture_W2L30_Lecture_8__Cloud_Computing__XML_Basics | Difficulty: Medium

Q27. Which XML parser is best suited for processing very large XML files or streaming data in real-time with minimal memory footprint, but does not allow for modifying the document's structure?

- A) DOM Parser
- B) SAX Parser
- C) XSLT Processor
- D) XPath Engine

Source: Lecture_W2L31_Lecture_9__Cloud_Computing__XML_Basics_II | Difficulty: Medium

Q28. In a Service-Oriented Architecture (SOA), which XML-based component acts as the 'user manual' or contract for a web service, describing its location, operations, and message formats?

- A) XML
- B) SOAP
- C) WSDL
- D) UDDI

Source: Lecture_W2L32_Lecture_10__Cloud_Computing__Web_Services__Service | Difficulty: Medium

Q29. A cloud service provider guarantees 99.95% availability for a service costing \$2000/month. The SLA specifies a 10% service credit if availability falls between 99.95% and 99.0%, and a 25% service credit if it falls below 99.0%. If the service experiences 10 hours of downtime in a 30-day month (assume 24 hours/day operation), what is the effective cost for the consumer that month?

- A) \$2000
- B) \$1800
- C) \$1500

D) \$1900

Source: Lecture_W3L37_Lecture_11__Service_Level_Agreement | Difficulty: Hard

Q30. An organization is evaluating whether to host a workload on-premise or use a cloud provider. Their workload has a Peak-to-Average demand ratio (P/A) of 5. The cloud provider charges a Utility Premium (U) of 3 (meaning cloud resources are 3 times more expensive per unit of time than on-premise). Based on the economic model discussed, which option is more economical, and why?

- A) On-premise, because the cloud is more expensive per unit.
- B) Cloud, because $P/A > U$, indicating the cost savings from elasticity outweigh the higher unit cost.
- C) On-premise, because the Utility Premium is greater than 1.
- D) Cloud, because the P/A ratio is high, regardless of the Utility Premium.

Source: Lecture_W3L38_Lecture_12__Cloud_Economics | Difficulty: Hard

Answer Key

1: B	2: B	3: C	4: B	5: D
6: B	7: C	8: C	9: C	10: B
11: D	12: B	13: B	14: B	15: B
16: C	17: C	18: B	19: C	20: C
21: B	22: C	23: B	24: B	25: B
26: A	27: B	28: C	29: C	30: B